

KOP Homework 01

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Executive Summary

ProbSAT beats GSAT on every metric (success rate, PAR10, flip count distribution).

1 Introduction

We set out to compare the two SAT algorithms, GSAT and ProbSAT, on our set of data. The better algorithm should converge faster, especially on larger instances.

2 Methods

2.1 Solvers

We used the GSAT implementation from Course Pages with a fixed parameter $p = 0.4$.

The probSAT solver was downloaded from Adrian Balint's GitHub and modified to return results in the same format as our GSAT implementation. Fixed parameters: $c_m = 0, c_b = 2.3$.

2.2 Instances

We chose to use the "ruf" set of instances from Course Pages. Specifically: ruf20-91, ruf36-157, ruf50-218, ruf75-320.

2.3 Experiment

To test the solvers, each solver was run 1 000 times per configuration. Each instance contains 1 000 configurations. In total, we tested 8 000 000 solver runs.

The max_flips parameter was set to 10 000 for all runs.

2.4 Measurements

For each experiment run, we saved the number of flips, whether the run was successful, the instance and configuration name, and the solver used. The cap was 10 000 flips, so unfinished runs always report 10 000 flips.

2.5 Implementation

The python implementation creates subprocesses which call the GSAT and ProbSAT binaries, which are run in parallel. It then aggregates all results into a single pandas dataframe. The dataframe is then saved and further processed in a marimo notebook to create the tables and figures.

To run the experiments, we used Metacentrum. The final experiment was run on a 32-core AMD Ryzen CPU, the run took roughly 1 hour.

2.6 Metrics

All tables and plots derive from the following metrics:

- **Success rate:** share of runs that satisfied all clauses.
- **PAR10:** penalized average runtime, calculated as mean flip count where each failed run is penalised by a factor of 10 ($10 \times 10\,000$ flips). This metric favours solvers that finish often and quickly.
- **Log-normal parameters:** mean and standard deviation of $\log(1 + \text{flips})$ across all runs, lower values imply faster and more concentrated performance. The metric makes sense under the assumption that the number of flips comes from a log-normal distribution.
- **Flip-count histogram:** visualization of flip-count distribution (both linear and log scale). Only successful runs are visualized because otherwise unsuccessful runs would saturate the histogram at the `max_flips` value and the actual flip count is unknown.
- **Empirical CDFs:** CDF of flips for successful runs, corrected "CDF" scaled by the success probability, making it reflect performance over all runs.

3 Results & Discussion

ProbSAT yields better results over all metrics.

Success rate (Table 1) is slightly in favor of ProbSAT, especially for the largest instance, where it clearly dominates.

ProbSAT yields significantly lower PAR10 scores (Table 2).

Both mean and standard deviation of $\log(1 + \text{flips})$ (Table 3, Table 4) are lower for ProbSAT, indicating faster convergence and greater stability.

The histogram of flip counts (Figure 1) shows that most of the problems are solved with a low number of flips, but it is too saturated to make any conclusions about which solver is better.

However if we look at the histogram of $\log(1 + \text{flips})$ (Figure 2), we see that it is consistent with the mean and std measured earlier. We can clearly see that ProbSAT is skewed left, towards lower flip counts.

In both CDFs (Figure 3, Figure 4), ProbSAT is above GSAT, indicating faster convergence.

4 Conclusion

ProbSAT is preferable to GSAT on the tested ruf instances.

Tables

Table 1: Success rate (%), higher is better.

dataset	GSAT	ProbSAT
ruf20-91	100.00	100.00
ruf36-157	99.20	99.93
ruf50-218	96.18	99.37
ruf75-320	85.96	96.55

Table 2: PAR10, lower is better.

dataset	GSAT	ProbSAT
ruf20-91	161.00	94.00
ruf36-157	1401.00	404.00
ruf50-218	4914.00	1296.00
ruf75-320	15670.00	4717.00

Table 3: Log-normal mean flips, lower is better.

dataset	GSAT	ProbSAT
ruf20-91	4.03	3.85
ruf36-157	5.36	5.01
ruf50-218	6.21	5.75
ruf75-320	7.15	6.59

Table 4: Log-normal std of flips, lower is better.

dataset	GSAT	ProbSAT
ruf20-91	1.35	1.13
ruf36-157	1.51	1.22
ruf50-218	1.54	1.25
ruf75-320	1.47	1.27

Figures

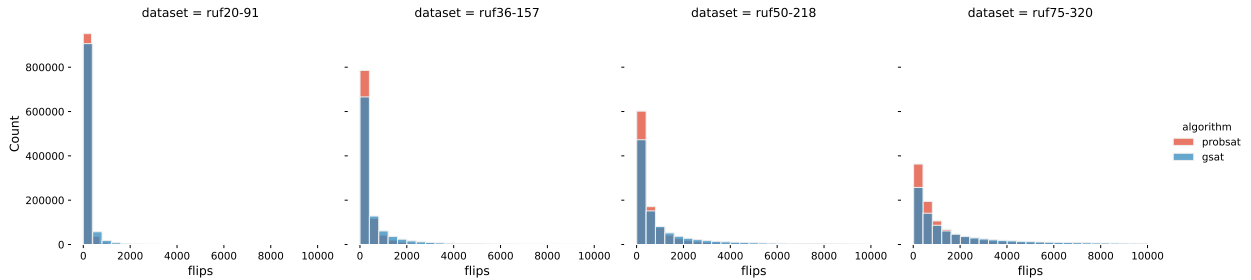


Figure 1: Histogram of flip counts for successful runs (linear scale).

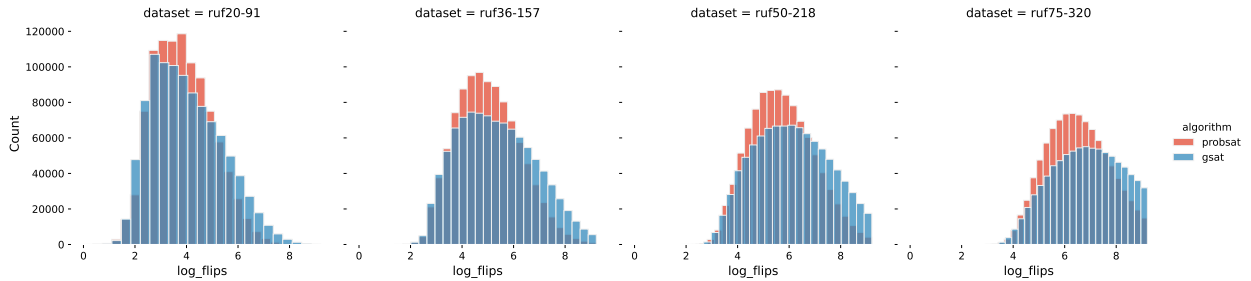


Figure 2: Histogram of flip counts for successful runs (log scale).

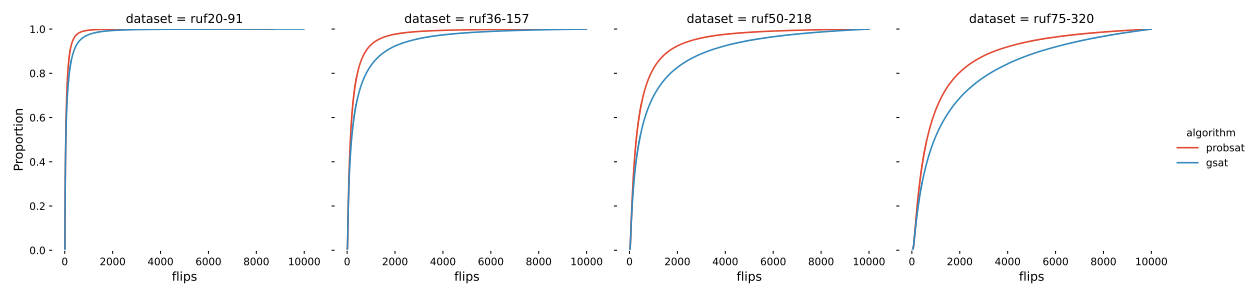


Figure 3: Empirical CDF of flip counts for successful runs only.

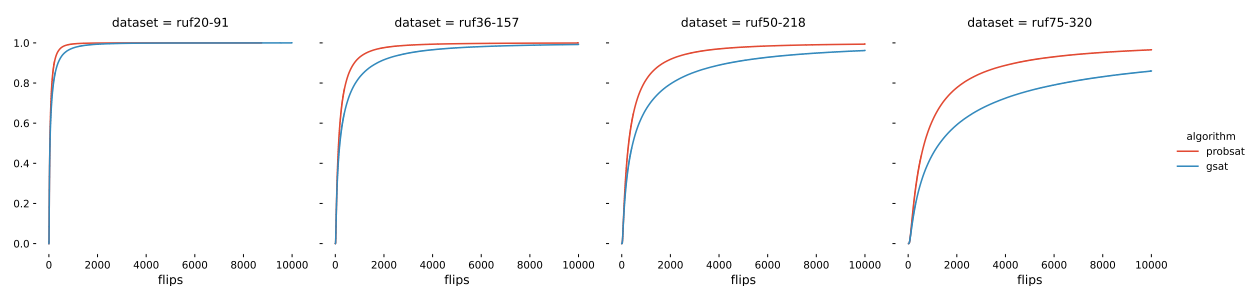


Figure 4: Empirical CDF of flip counts over all runs (success-weighted correction for unfinished runs).